From DNA Extraction to PCR Verification

Minor Project · Jayosi Gayen · Oct 26, 2026

Introduction

Aim: To understand how a researcher obtains DNA, checks its quality, amplifies a target region and evaluates the result, linking DNA extraction, quantification, PCR, gel electrophoresis and sequencing.

Each stage answers one question. Do I have DNA? Is it pure enough? Did I copy the right region? Is the product the expected size? Is the sequence actually correct?

1. DNA extraction

Why extract? In a cell, DNA is mixed with proteins, lipids, RNA and nucleases. These can degrade DNA or inhibit the PCR enzyme, so DNA must be isolated in a clean, stable form before it can be measured or amplified.

Workflow: Sample → Break cells → Remove contaminants → Purify molecule → Check quality → Analyze

1. **Break cells (lysis):** detergent (SDS) dissolves membranes; EDTA blocks DNases.
2. **Remove contaminants:** proteinase K digests proteins, RNase removes RNA, and debris is spun down.
3. **Purify:** DNA is precipitated with cold ethanol (or bound to a silica column), washed and dissolved in TE buffer.

2. DNA quality check (quantification)

A UV spectrophotometer measures how much DNA is present and how pure it is.

Reading	Measures	Purpose
A260	Nucleic acids	DNA concentration (A260 of 1.0 = about 50 µg/mL dsDNA)
A280	Proteins	Detects protein contamination
A260/A280	Ratio	Purity: about 1.8 = pure; below 1.7 = protein; above 2.0 = RNA

Impure DNA can make PCR fail, and knowing the concentration lets the right amount of template be added.

3. Polymerase chain reaction (PCR)

PCR copies one target region millions of times, using primers, heat-stable Taq polymerase and dNTPs. Each cycle has three stages:

1. **Denaturation (94–98 °C):** heat separates double-stranded DNA into single strands.
2. **Annealing (50–65 °C):** primers bind to their complementary sites flanking the target.
3. **Extension (72 °C):** Taq polymerase adds dNTPs to the primers, building new strands.

After 25–35 cycles the target has doubled each time, giving enough copies to see on a gel.

4. Agarose gel electrophoresis

PCR products are loaded into an agarose gel and an electric current is applied. Negatively charged DNA moves toward the positive electrode, and smaller fragments travel farther. A stain makes the bands visible under UV light.

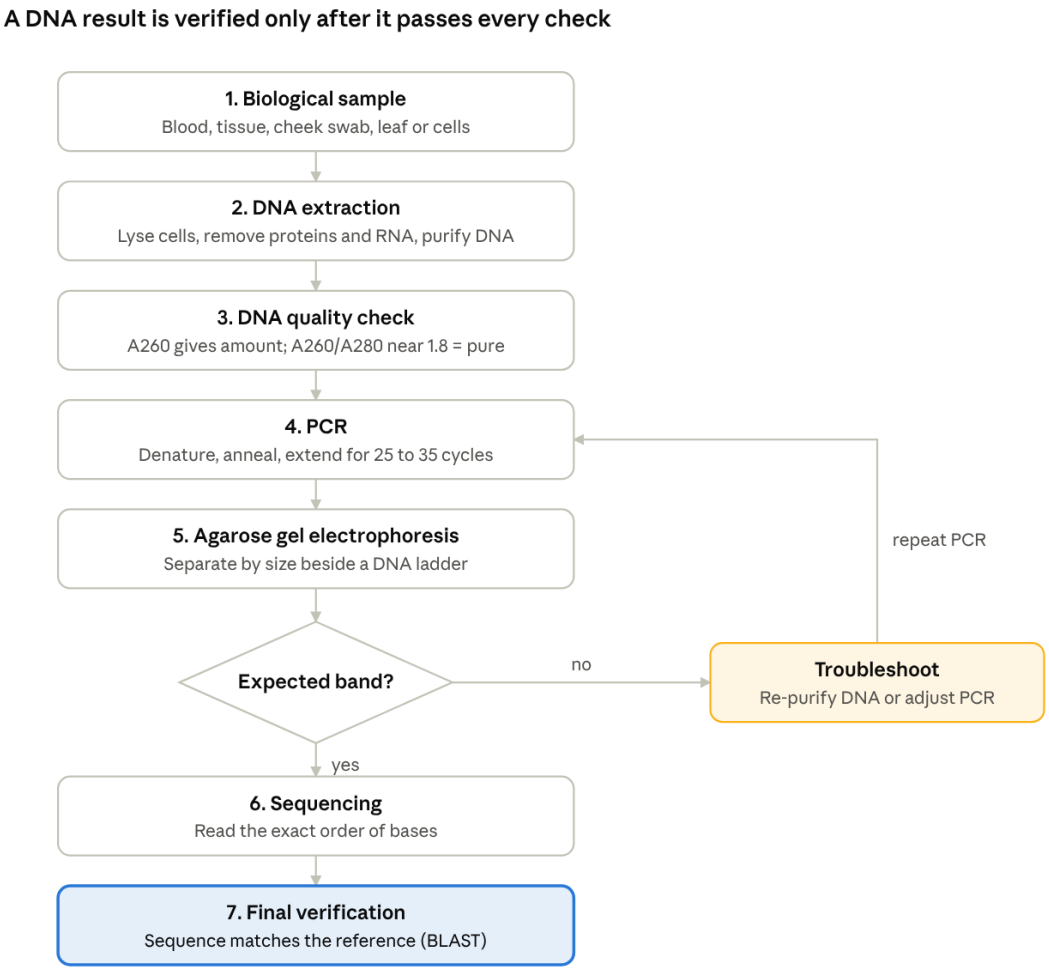
- **One sharp band at the expected size:** successful PCR.
- **No band, extra bands or primer-dimers:** failed or non-specific PCR.

Why a DNA ladder? The ladder holds fragments of known sizes run beside the samples. Comparing the sample band with it gives the product's size in base pairs, confirming it matches the expected length.

5. Sequencing for final confirmation

A gel shows only size, and different DNA fragments can be the same length. It also cannot reveal point mutations or small changes. Sequencing (usually Sanger) reads the exact base order of the PCR product, which is then compared with a reference sequence using BLAST. A close match confirms the correct target was amplified.

6. Labelled workflow



Labelled workflow · sample to verified sequence

If the gel shows no band, the wrong size or extra bands, the researcher goes back, re-checks the DNA or adjusts the PCR, and runs it again; only a correct band moves on to sequencing.

Conclusion

Each stage protects the next: extraction gives clean DNA, the quality check confirms it is usable, PCR amplifies the target, the gel checks its size, and sequencing proves its identity. Skipping any step makes the final result unreliable; only when all five agree is the result verified.